

Healthcare Appointment Attendance Analytics

1. Project Overview

1.1 Project Title

Healthcare Appointment Attendance Analytics: Identifying Factors Driving Patient No-Shows in Healthcare Appointments

1.2 Business Problem

Healthcare providers lose revenue, waste staff resources, and reduce operational efficiency when patients fail to attend scheduled appointments. This project analyzes more than 110,000 healthcare appointments to identify the key factors influencing patient no-show behavior and provide actionable recommendations.

1.3 Business Objective

Determine how age, gender, neighborhood, scholarship status, hypertension, diabetes, alcoholism, handicap, SMS reminders, and appointment waiting time influence appointment attendance.

1.4 Project Goal

Identify the primary drivers of patient no-shows and provide data-driven recommendations that healthcare providers can use to improve appointment attendance rates and operational efficiency.

1.5 Technology Stack

PostgreSQL, SQL, Python (Pandas, NumPy, Matplotlib, Seaborn) in Jupyter Notebook, Power BI (Windows VM on Mac), VS Code, Git, and GitHub.

2. Project Architecture

2.1 Architecture

Kaggle CSV → Data Cleaning → PostgreSQL Database → Star Schema → SQL Analysis → Python EDA
→ Power BI Dashboard → GitHub Portfolio

2.2 Data Model

A dimensional data model (Star Schema) will be implemented to support analytical reporting.

Fact Table:

- fact_appointments

Dimension Tables:

- dim_patient
- dim_health_condition
- dim_date

The model separates transactional appointment data from descriptive patient and health attributes, enabling efficient reporting, KPI analysis, and dashboard development.

2.3 Technical Skills Demonstrated

Data Modeling, SQL Joins, CTEs, Window Functions, Views, Data Cleaning, Exploratory Data Analysis (EDA), KPI Development, Dashboarding, Business Analysis, and Data-Driven Recommendations.

2.4 Key Performance Indicators (KPIs)

- Total Appointments
- Attendance Rate
- No-Show Rate
- Average Appointment Waiting Time
- No-Show Rate by Age Group
- No-Show Rate by Gender
- No-Show Rate by Neighborhood
- SMS Reminder Effectiveness
- High-Risk Patient Segments

3. Final Deliverables

- PostgreSQL Database with Star Schema
- SQL Analysis Scripts
- Python EDA Notebook (Jupyter)
- Power BI Dashboard
- GitHub Portfolio Repository
- Project Documentation and Business Recommendations

4. GitHub Development Strategy

The project will follow a version-controlled development approach. Every major milestone will be committed and pushed to GitHub, including project setup, data profiling, database design, SQL development, Python EDA, Power BI dashboard creation, documentation, and final project delivery.